

BERKAY KULLUKÇU

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EDUCATION

- Doctor of Engineering (Dr. Ing.) in Electrical Engineering** 2023 - ...
Technische Universität Dresden, Chair of Acoustics and Haptics, Dresden, Germany.
- Master of Science (MSc) in Mechanical Engineering** 2020 - 2022
Koç University, College of Engineering, Istanbul, Turkey.
- Bachelor of Science (BSc) in Mechanical Engineering** 2015 - 2020
Koç University, College of Engineering, Istanbul, Turkey.

SKILLS AND INTERESTS

- Acoustics & Psychoacoustics:** Electric Trucks, Vehicle Acoustics, AVAS, Sound Quality, Semantic Descriptor Modeling, Psychoacoustic Metrics (e.g., Roughness, Tonality, Sharpness), Field Studies, Project Proposal Writing (DFG)
- Signal Processing & Simulation:** SQL, MATLAB, Python, COMSOL FEM, Artemis Suite, Data science, Statistics, Machine Learning, Object-oriented programming
- Measurement & Experimentation:** Free-field Microphones, Accelerometers, Squadriga II/III, Binaural Recording, Listening Tests, Perceptual Modeling
- Languages:** English (C2), German (C1)

RESEARCH EXPERIENCE

- Department of Machine Dynamics and Acoustics (Scientific Employee) (TH Wildau) - Wildau, Germany** October 2025 - ...
- Conducting research on acoustic and vibration-based condition monitoring of rail vehicle components, with a focus on detection of loose bolted joints and structural defects.
- Chair of Acoustics and Haptics (Dr. Ing.) (AHA) - Dresden, Germany** April 2023 - ...
- Conducted research on AVAS sound design for electric trucks including psychoacoustic testing, semantic evaluation, and binaural sound synthesis. Experiments designed to measure user comfort, and perceptual urgency of vehicle warning sounds.
- Bio-Integrated Microdevices Laboratory (Master's) (BMDL) - Istanbul, Turkey** June 2020 - September 2022
- Research on flexible polymer energy harvester, PMUTs, nano/microfabrication, acoustic equivalent circuit models.
- Bio-Integrated Microdevices Laboratory (Undergraduate) (BMDL) - Istanbul, Turkey** Jan 2020 - June 2020
- Research on wearable sensor system design for venous dislodgement detection in hospital patients.

TEACHING EXPERIENCE

- TH Wildau - Wildau, Germany** Oct 2025 - Jan 2026
- Conducted the teaching assistant duty in the senior engineering class, IDM (Interdisciplinary Module).
- TU Dresden - Dresden, Germany** June 2024 - Jan 2025
- Supervised a Master's (Diplomarbeit) student for a 6 month vehicle acoustics (Forschungspraktikum) internship (Student's class note: 1.0).
- TU Dresden - Dresden, Germany** September 2023 - ...
- Vorlesung Sound Design - Supervising senior engineering students for a vehicle sound design project.
 - Acoustic Hands-on Experimenting (Praktikum) Class - Supervising senior engineering students on a household appliance sound power measurement project.
- Koç University - Istanbul, Turkey** September 2020 - Jan 2021
- MECH401 - "Mechanical Engineering Laboratory" tutoring to the senior engineering class.

The Educational Volunteers Foundation of Turkey (TEGV) - Istanbul, Turkey

Sep 2018 - Jan 2019

- Programming tutor for a class of middle school children.

Mathematics Teacher - Istanbul, Turkey

2017- 2020

- Private mathematics tutoring to primary, middle school children.

INDUSTRIAL EXPERIENCE

GENERAL ELECTRIC (GE) AEROSPACE - Kocaeli, Turkey

Nov 2022 - March 2023

Design Engineer

- Design engineer on CT7/T700 turboshaft & turboprobe engines.

TEDAVEM Information Systems (LLC) - Istanbul, Turkey

June 2021 - Jan 2023

Co-Founder

- Start-up working on hospital digitalization, IoT systems for monitoring ventilation, infusion machines in patient rooms. Integrated circuit design for device communication.

KORDSA - Istanbul, Turkey

March 2022 - September 2022

Flexible Electronics Researcher/ R&D

- Thin Films& Flexible Electronic Sensors Department (0.35% applicant acceptance rate).
- Labview software development for testing.

TURKISH AEROSPACE (TAI) - Ankara, Turkey

November 2019 - May 2020

Mechanical Engineering Internship

- Safety Management Systems Department engineer.

GROUPE RENAULT - Bursa, Turkey

June 2019 - August 2019

Automotive Engineering Internship/ R&D

- Software development for helping R&D engineers manipulating the 3D parts in Catia (20k+ lines of code).

TURKISH AEROSPACE (TAI) - Istanbul, Turkey

November 2019 - January 2020

Mechanical Engineering Internship/ R&D

- Patran FEM and 3D CAD work for aerospace vehicle part design.

PUBLICATIONS

1. B. Kullukçu, J. Krautwurm, S. Merchel, R. Rosenkranz, and M. E. Altinsoy, "Investigating perceptual discrimination thresholds for attributes of whole-body vibration," *Scientific Reports*, 2026.
[Link](#)
2. B. Kullukçu, R. Rosenkranz, and M. E. Altinsoy, "Electric Truck Sound Design: A User-Centered Psychoacoustic Approach to AVAS," *Journal of the Audio Engineering Society*, 2025.
[Link](#)
3. B. Kullukçu et al., "Piezoelectric PVDF-TrFE/PET Energy Harvesters for Structural Health Monitoring (SHM) Applications," *Integrated Ferroelectrics*, 2023.
[Link](#)
4. B. Kullukçu, and L. Beker, "Design and Modeling of a PVDF-TrFe Flexible Wind Energy Harvester," *Turkish Journal of Electrical Engineering and Computer Sciences*, 2023.
[Link](#)
5. Piezoelectric PVDF-TrFE PET Energy Harvesters for Structural Health Monitoring (SHM) Applications in Wind Turbines - Master's Thesis, 2023.
[Link](#)

CONFERENCES

1. B. Kullukçu, S. Atamer, J. Krautwurm and M.E. Altinsoy, "Perceived Annoyance in Multi-Source Electric Vehicle AVAS Environments," in *ICSV32, Istanbul, Turkey, 2026 - Accepted*.

2. B. Kullukçu, R. Pianowski, and D. Hannebauer, "Frequency-Modulated and Single-Tone Excitation to Reveal Vibro-Acoustic Nonlinearities in Loosened Bolted Joints," in Forum Acusticum, Graz, Austria, 2026 - Accepted.
[Link](#)
3. B. Kullukçu, R. Kamenzky, and D. Hannebauer, "SIAMIS: Experimental Study on the Acoustic Behavior of Bolted Joints Under Varying Tightening Torques," in Proc. DAGA, Dresden, Germany, 2026.
[Link](#)
4. B. Kullukçu, M. Engel, S. Atamer, and M.E. Altinsoy, "Psychoacoustic Modelling of AVAS Sounds: Consumer-Centric Semantic Attribute Development for Electric Vehicles," in Proc. Forum Acusticum, Málaga, Spain, 2025.
[Link](#)
5. B. Kullukçu, O. Zeidler, S. Atamer, and M. E. Altinsoy, "Experimental Determination of the Masking Threshold in Electric Trucks," in Proc. DAGA, Copenhagen, Denmark, 2025.
[Link](#)
6. B. Kullukçu, J. W. Lee, S. Atamer, and M. E. Altinsoy, "Designing Effective Warning Sounds: Psychoacoustical Considerations," in Proc. INTER-NOISE, Nantes, France, 2024.
[Link](#)
7. B. Kullukçu, R. Rosenkranz, and M. E. Altinsoy, "Acoustic Vehicle Alerting Systems (AVAS) for Electric Trucks: Initial Thoughts for the Requirements," in Proc. DAGA, Hannover, Germany, 2024.
[Link](#)
8. P. T. Ziarati, B. Kullukçu and L. Beker, "Modeling miniaturized piezoelectric ultrasound transducers: comparison of lumped and finite element models," in 2020 IEEE International Ultrasonics Symposium (IUS), Las Vegas, NV, USA, 2020.
[Link](#)

PATENTS

1. B. Kullukçu, G. Umut, and B. Ataseven, "Momentum Nasogastric Tube System," TR2022001302A1, Oct. 21, 2024. - [Link](#)
2. B. Kullukçu, G. Umut, and B. Ataseven, "An Invasive Device," TR2021015887A1, May. 21, 2024. - [Link](#)
3. B. Kullukçu, G. Umut, and B. Ataseven, "Detecting Device and System For Infusion Pumps," Utility Model TR2022007637Y, Apr. 22, 2024. - [Link](#)

SCHOLARSHIPS & AWARDS

1. Travel Grant for ICSV32 (Förderprogramm Kongressreisen 2026) - DAAD
2. EAA Travel Grant for Forum Acusticum 2025 Conference - European Acoustics Association
3. V. Patent Day (2019-2021) Patent Holder Award - Koç University
4. BİDEB 2210-C "National Scholarship Program for MSc Students (Priority Research Areas)" Scholar - TÜBİTAK
5. TEYDEB 1512 - 25000\$ Techno-Startup Funding - TÜBİTAK
6. Design Innovation Challenge 1st Prize - Undergraduate & Graduate Scholarship - Garip & Zeycan Foundation
7. Graduate Academy Conference Travel Grant for DAGA 2025 Copenhagen - TU Dresden
8. VI. Patent Day Patent Holder Award - Koç University

SESSION CHAIR DUTIES

1. "Körperschall" (Body-induced Noise) Session - DAGA 2026, Dresden, Germany - [Link](#)

2. T14 SS03: Sound Quality and Product Design - 32nd International Conference on Sound and Vibration (ICSV32) 2026, Istanbul, Turkey - [Link](#)

REFERENCES

1. Prof. Dr.-Ing. Ercan Altinsoy
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ercan.altinsoy@tu-dresden.de
2. Dr. Levent Beker
Asst. Professor, Koç University
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